

Nonlinear propagation effects in an AlGaAs Bragg grating filter

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We present experimental observations of nonlinear propagation effects in an integrated AlGaAs waveguide filter. We demonstrate pulse shaping, pulse compression, and the production of gap solitons within the stop band of the grating for switching powers of ~ 130 W. This nonlinear behavior is associated with the counterbalancing effects of self-phase modulation and the large dispersion effects introduced by the grating itself. © 1999 Optical Society of America

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Recent theoretical and experimental work suggests that integrated nonlinear grating filters have potential uses in future high-bandwidth telecommunications and optical computing systems. They can be used to process optical pulses in a variety of ways, such as filtering, shaping, multiplexing, optical power limiting, and all-optical switching. These nonlinear properties have been investigated extensively since Winful and co-workers theoretically predicted bistability, multistability,¹ chaos,² switching, and pulse compression³ in periodic structures. In general, the ideal nonlinear grating filter should have low switching power, high switching speed, high throughput, and high on-off contrast. Furthermore, the device should be compact and easily fabricated.

To date, nonlinear propagation effects in grating structures have been experimentally investigated in several material systems. For example, slow (nanosecond response time) thermally induced nonlinear grating effects were explored within colloidal crystals⁴ and in InSb (Ref. 5) planar structures at wavelengths close to the central Bragg wavelength. Nonlinear switching was also demonstrated within integrated semiconductor waveguide filters that exhibit resonant nonlinearities, including optical switching in a silicon-on-insulator waveguide at $1.064\text{ }\mu\text{m}$ by use of switching energies of ~ 1 to $18\text{ }\mu\text{J/pulse}$.⁶ More recently, contradirectional coupling and pulse routing by use of a two-pulse interaction mechanism was demonstrated based on the photogenerated-carrier-induced nonlinearity of InGaAs/InP multiple quantum wells.^{7,8} All-optical switching,⁹ logic gate operation,¹⁰ and soliton production^{11,12} were also achieved in fiber Bragg gratings.

In this Letter we report the experimental observation of the formation of gap solitons within an AlGaAs integrated Bragg waveguide. AlGaAs ridge waveguides are ideal structures for investigating the non-

linear optical properties of gratings, as they provide good optical confinement over large interaction lengths, with low linear losses of approximately 0.7 to 2 dB/cm. Moreover, large effective index modulations, Δn_{eff} , as great as 0.1, can be achieved in such structures. The AlGaAs guides also have sufficient power-handling capabilities to perform these types of nonlinear experiment. For example, in this experiment incident peak powers of up to 1.2 kW (660 W when coupling losses are considered) were launched into the guides without optical damage's occurring. In contrast with the integrated semiconductor grating experiments discussed in Refs. 6–8, our device was based on nonresonant nonlinearity induced at an operating wavelength below the half-bandgap region of AlGaAs, i.e., at the $1.55\text{-}\mu\text{m}$ low-loss telecommunications window. At this wavelength the detrimental effects of two- and three-photon absorption were minimized through wafer design, and the nonlinear refractive index n_2 was measured to be $\sim 1.5 \times 10^{-13}\text{ cm}^2/\text{W}$.¹³ Therefore the switching power required for such devices is considerably lower than that reported for similar structures in fiber, as discussed in Refs. 9–12. Since the nonlinearity is based on a nonresonant process, the material response time is almost instantaneous (femtoseconds), and the performance of the device is not limited by linear absorption effects, as is the case for many devices based on resonant nonlinearities. Fabrication of integrated semiconductor waveguide filters has commonly been achieved with a two-step process in which the waveguides are defined prior to formation of the grating.^{6–8} In our experiment the grating and the waveguide were defined simultaneously by use of a simple one-step electron-beam-based approach.¹⁴ We based the filters on a weak grating on a strip-loaded waveguide to minimize excess scattering losses. In this way, 4–6-mm-long high-quality grating filters with low scattering losses of 2 dB/cm were produced. This electron-beam

fabrication process was subsequently used to fabricate more-complex grating filters, including chirped and apodized filters.¹⁵

The AlGaAs wafer used to fabricate the waveguide-grating filters was grown by molecular beam epitaxy and had the following structure: The lower cladding layer was 4 μm thick and contained 24% Al, the waveguide layer was 1.5 μm thick and contained 18% Al, and the upper cladding was 1 μm thick with 24% Al. Both 4- and 6-mm-long gratings were written on 1-cm-long guides that were designed to be single mode. The width of the guides was chosen to be 5 μm , resulting in an effective mode area of $\sim 12 \mu\text{m}^2$. We selected a grating period of 235 nm to position the grating stop band around the maximum power range of the laser at ~ 1533 nm. The waveguides were etched to a depth of 0.95 μm . Owing to reactive-ion etching lag, the grating etch depth was $\sim 0.3 \mu\text{m}$, resulting in an effective index modulation depth of $\sim 4.4 \times 10^{-4}$. Figure 1 shows a schematic of the grating filter design and also includes a scanning electron microscope photo of an actual device. We removed the waveguide section prior to our experiments to eliminate any self-phase modulation effects occurring outside the grating itself. The length of the grating used in this experiment was 4 mm, and the grating had a maximum reflectivity of $\sim 97\%$ at 1531.76 nm ($\kappa L \sim 2.3$) and a stop-band width of 0.4 nm (see Fig. 2).

High-power high-quality pulses with narrow spectral bandwidths are required for observation of nonlinear effects within the grating filters. Therefore we used pulses from an externally modulated distributed-feedback laser amplified by a large-mode-area erbium-doped fiber amplifier cascade,¹⁶ which generated 400–500-ps rectangular pulses with repetition rates of 10–100 kHz. Pulses with spectral bandwidths of ~ 3 GHz and peak powers of up to 10 kW could readily be achieved with this system. The wavelength of the pulse could be varied from 1.52 to 1.56 μm .

The pulses were coupled into the waveguide by a $40\times$ objective. After propagation, the transmitted light was coupled out of the guide by another $40\times$ lens. We then steered the output pulses alternately into a spectrum analyzer and a fast optical detector-sampling scope via an optical fiber to measure the spectral and temporal characteristics of the transmitted pulses. The average input power was monitored before the input coupling lens by use of a pyroelectric meter, and the pulse peak power was evaluated with measurements from a fast photodiode and an oscilloscope. A polarizer-half-wave-plate combination was used to select the input polarization, and a second wave plate, placed before the polarizer, was used to control the amount of light coupled into the guides. It should be noted that in the following experiments we estimated the peak input power of the pulse by taking into consideration that the losses associated with coupling into the guide by use of the objective lens amounted to 55% of the input peak power, including the 30% loss owing to the reflectivity of the AlGaAs front facet.

Nonlinear pulse propagation effects were explored within the stop band of the grating filters. Initially, as a reference, we tuned a low-power input pulse to

lie outside the stop band of the grating, on the short-wavelength side, at 1530.46 nm, where the effects of the grating are negligible. The FWHM of the output pulse from the grating was measured to be 405 ps for a peak input power of ~ 10 W. The pulse shape and spectrum were then monitored as the input power was increased, and no significant nonlinear distortion of the pulse was observed. This implies that we can neglect the effects of pure self-phase modulation in discussing the results for frequencies within the grating's bandwidth. The low-power, input pulse was then tuned to the central reflecting wavelength of the grating at 1531.76 nm; Fig. 2 shows the position of the pulse with respect to the grating stop band. The input pulse power was then increased, and the temporal characteristics of the output pulse were measured for several input powers, as shown in Fig. 3. The input peak power, P_{in} , and the output pulse width at FWHM, T_p , are indicated for each measurement. The FWHM output pulse width is 435 ps, which is 30 ps broader than the pulse tuned outside the gap. Figure 3 also shows the output pulse at higher intensities, indicating that pulse shaping and significant pulse compression

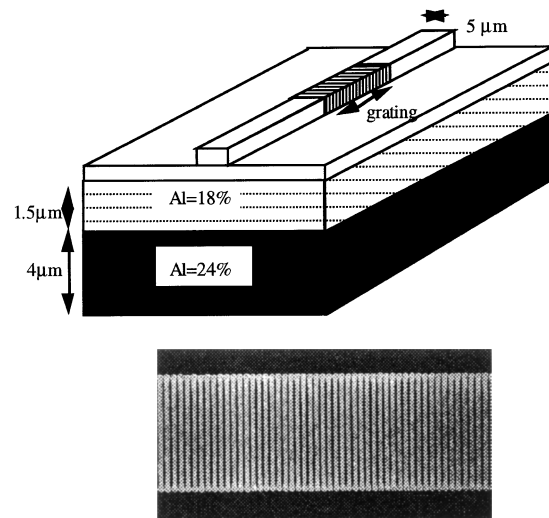


Fig. 1. Top, schematic of the integrated AlGaAs waveguide; bottom, scanning electron microscope photograph showing the top view of a grating filter with a 0.235- μm -period grating on a 5- μm -wide waveguide.

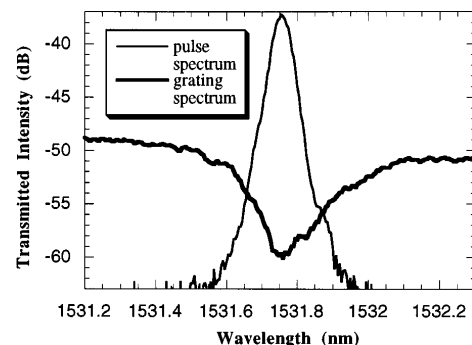


Fig. 2. Position of the input pulse with respect to the grating stop band.

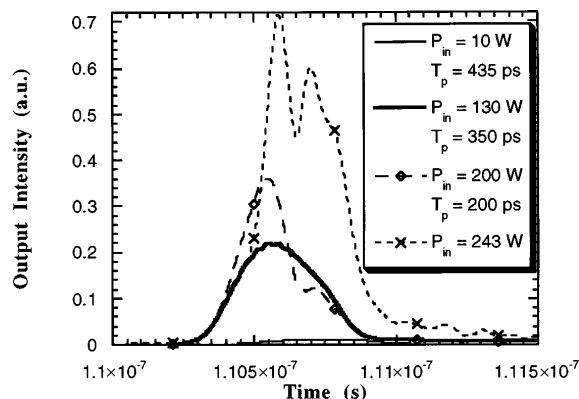


Fig. 3. Output temporal pulse profiles as a function of input peak power for increasing input power.

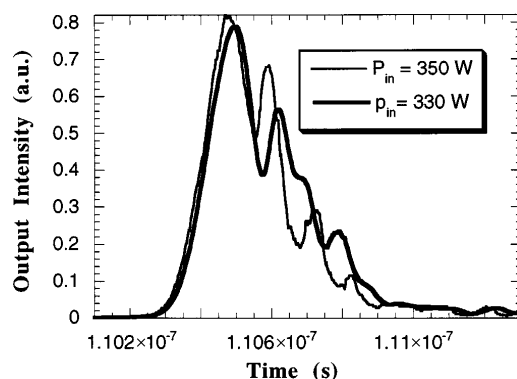


Fig. 4. Multiple soliton production at high input peak powers.

occur as the input pulse power is increased. It is obvious from these figures that true pulse compression as opposed to truncation occurs at the center of the gap, as there is an enhancement in the peak power of the pulse at high input pulse powers. At an input peak power of 200 W the pulse has been compressed by a factor of 2.1, and pulse splitting is evident. The division of the input pulse into two pulses is more obvious for an input pulse power of ~ 243 W. This behavior is due to the effects of the grating, since high-power pulses tuned outside the grating do not show this effect. The pulse compression shown in these figures is associated with the formation of a single gap soliton, which in this instance occurred for a power of approximately 130 W. At even higher input pulse powers, as shown in Fig. 4, the grating emitted a train of periodically spaced pulses. This pulse-splitting behavior has been theoretically associated with modulational instabilities within the grating¹⁷ and has been experimentally demonstrated in fiber gratings.¹⁸ This effect could be used for short-pulse generation. Overall, we found that there was a 10-fold nonlinear increase in the transmission at high powers when the pulse was initially tuned to the center of the gap. We also measured a 20-fold nonlinear increase in transmission at high powers for light detuned from the

Bragg resonance on the short-wavelength side but still within the stop band of the grating.

In conclusion, we have presented experimental results describing the nonlinear optical properties of an integrated AlGaAs Bragg waveguide filter. We have demonstrated nonlinear pulse compression and pulse shaping. In addition, the generation of multiple gap solitons was observed for switching powers as low as 130 W ($I \sim 1$ GW/cm²). These nonlinear effects were attributed to the interaction between the nonlinearity of the AlGaAs material and the large dispersive effects introduced by the grating. Numerical modeling of the system indicates that one can significantly increase the modulation depth of the pulse train in Fig. 4 by increasing the length of the grating. We plan to investigate the nonlinear properties of various chirped and deeply etched grating structures in AlGaAs and will aim to reduce the switching powers of these devices by optimizing the filter structure.

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